

Perceived Neighborhood Safety Effects on Adolescent Summer Activity

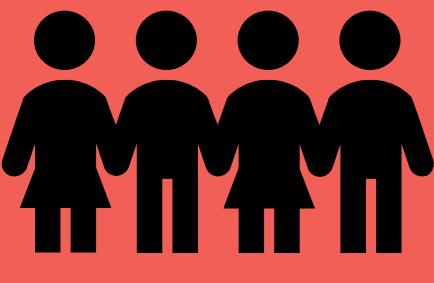


Sreemushta S.,¹ Park J.-H.,¹ Croan V.,¹ Dunton G.,² de la Haye K.,³ Pollack Porter K.,⁴ Lee C.,¹ Prochnow T.¹
1-Texas A&M University, 2- University of Southern California, 3- UCLA, 4 – Johns Hopkins University

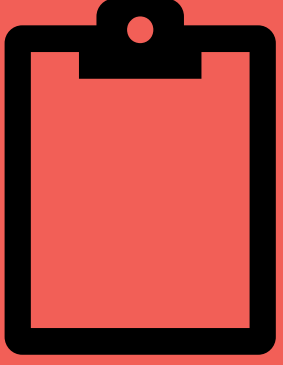
Introduction

- Adolescence is a critical period for health behavior development like physical activity (PA).
- Many adolescents fail to meet PA recommendations during summer.
- Aspects of the neighborhood environment like safety and aesthetics are significantly associated with PA.
- This study examined associations between neighborhood environment (aesthetics, pedestrian traffic safety, and crime safety) and multiple PA outcomes among adolescents during the summer.

Methods



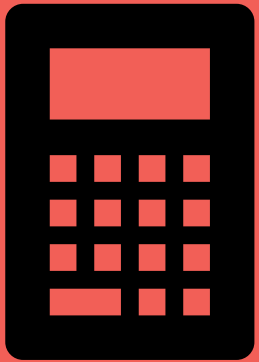
Youth aged 11 to 15 were recruited in summer 2025 through school districts in Central Texas (n=41).



The Neighborhood Environment Walkability Scale assessed perceived aesthetics, pedestrian safety, and crime safety.



PA was measured using wrist-worn ActiGraph LEAPS for 7 days, capturing intensity and total activity minutes.



Multilevel models examined associations between neighborhood perceptions and: (1) daily PA, (2) meeting daily guidelines (≥60 minutes moderate-to-vigorous PA (MVPA)), and (3) PA consistency scores, controlling for day in study, age, and sex.

Results

Table 1: Associations Between Neighborhood Environment and Physical Activity Separated by Intensity

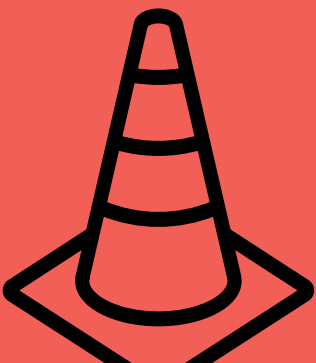
Variable	Light PA	Moderate PA	Vigorous PA
Neighborhood Environment			
Aesthetics	-2.79 (p=.80)	0.76 (p=.83)	-0.14 (p=.88)
Pedestrian Traffic Safety	-0.92 (p=.95)	-1.83 (p=.72)	-1.74 (p=.20)
Crime Safety	-4.22 (p=.66)	2.12 (p=.48)	1.88 (p=.02)*
Demographics			
Age	13.89 (p=.08)	5.11 (p=.04)*	0.21 (p=.75)
Female	41.52 (p=.007)**	10.98 (p=.02)*	-0.28 (p=.82)

Table 2: Associations Between Neighborhood Environment and Total Physical Activity, Meeting PA Guidelines, and PA Consistency

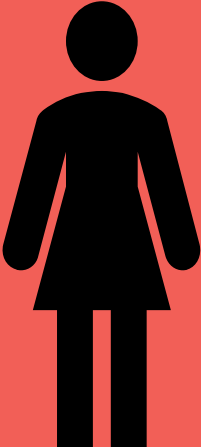
Variable	Total PA	Meets PA Guidelines	PA Consistency
Neighborhood Environment			
Aesthetics	-2.23 (p=.87)	OR=1.78 (p=.08)	-0.03 (p=.46)
Pedestrian Traffic Safety	-4.53 (p=.82)	OR=0.40 (p=.047)*	-0.09 (p=.16)
Crime Safety	-0.25 (p=.98)	OR=2.29 (p=.001)**	0.09 (p=.019)*
Demographics			
Age	19.26 (p=.04)*	OR=1.54 (p=.05)	0.01 (p=.70)
Female	52.18 (p=.006)**	OR=2.25 (p=.03)*	-0.04 (p=.48)



Perceptions of greater crime safety were associated with MVPA (B=1.88, p=.02), greater odds of meeting daily PA goals (OR = 2.29, p=.001) and more consistent PA (B= 0.09, p=.02).



Pedestrian traffic safety negatively predicted meeting guidelines (OR = 0.40, p = .04).



Girls were significantly more physically active and more likely to meet guidelines (OR = 2.25, P = .03).



Age was positively associated with moderate and total PA.

Conclusions



Adolescents who perceive their neighborhoods as safer are more likely to meet daily PA guidelines and have more consistent PA.



Improvements in crime safety in neighborhoods may be an effective strategy to promote PA in youth.

